UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

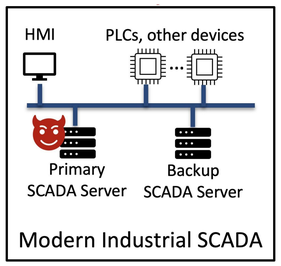
NATIONAL ENERGY TECHNOLOGY LABORATORY

From Legacy to Resilient: Digital Twin-Based Transition for Industrial Control Systems

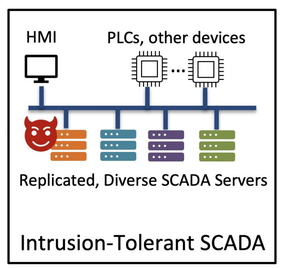
Huzaifah Nadeem [1], Amy Babay [1,2]
[1] Department of Computer Science
[2] Department of Informatics and Networked Systems

INTRODUCTION

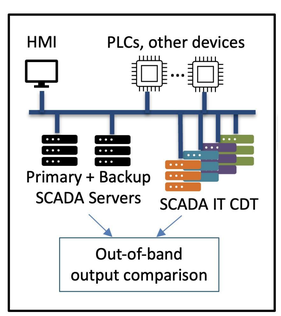
Today's SCADA systems are vulnerable to cyberattacks



Prior work shows how to build more resilient systems



Intrusion-Tolerant CDTs offer a path to transition.
Now, as well as in the future for emerging threats

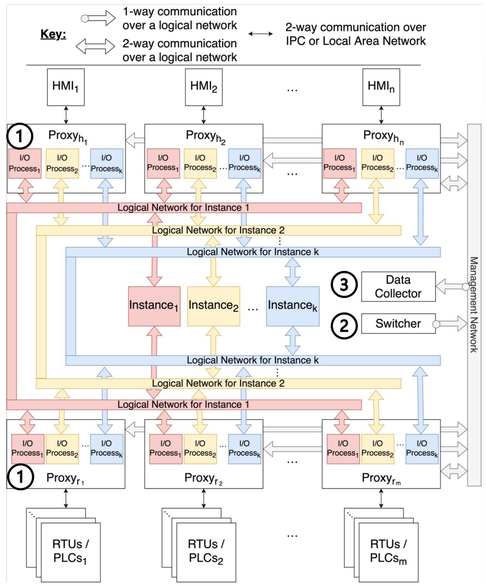


Out-of-band output comparison supports *intrusion detection* in the near term

ARCHITECTURE FOR SCADA SYSTEMS WITH CYBER DIGITAL TWINS

We can run any number of system instances out of which only one is the *active* instance. The rest are the *twins.*

The instances can differ in terms of the configurations they use, code-level diversity, and even the communication protocols they use.



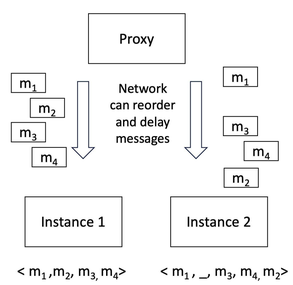
① I/O Processes completely isolate the instances. They allow us to translate any protocols differences as well as to dynamically add or remove any of the instances.

② The operator can seamlessly transfer control between the instances by using the Switcher.

③ The Data Collector logs all the system instances' events based on intrusions can be detected. The logs also help the operators evaluate the instances to build confidence when they upgrade the system.

SYSTEM INSTANCES OUTPUT COMPARISON

- Running an intrusion-tolerant digital twin alongside the active system produces parallel output streams that should be equivalent under correct operation.
- Any divergence between them could be a signal of compromise or malfunction.
- The technical challenge is that wide-area message reordering causes benign differences; we apply, at the data collector, runtime monitoring for safety properties to distinguish these from genuine anomalies.

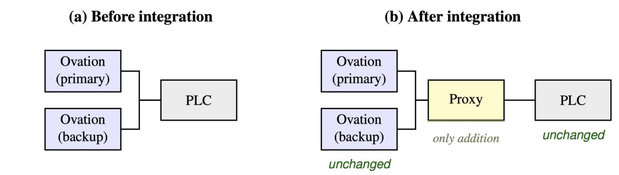


TEST DEPLOYMENT

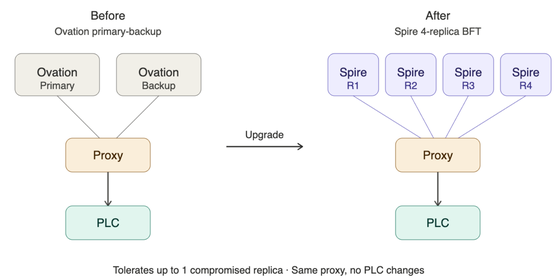
We validated our framework's generality by upgrading a control system built on Emerson Ovation, a widely deployed commercial controller in the power industry.

Test Setup
- Two Ovation OCR3000 controllers (primary-backup)
- Virtual PLC via OpenPLC on a low-power Linux machine
- Modbus TCP for controller–PLC communication

Plug-and-Play Integration A proxy machine was inserted in-line between the Ovation controllers and PLC — no IP changes or reconfiguration needed on either side. Integration caused minimal interruption to existing components.



From Primary-Backup to Intrusion-Tolerant Control
Building on this integration, we upgraded the Ovation primary-backup deployment to an intrusion-tolerant configuration using Spire, an open-source SCADA system, running a 4-replica Byzantine Fault Tolerant (BFT) setup.



ACKNOWLEDGEMENT

This material is based upon work supported by the Department of Energy under Award Number DE-CR0000039 and Pitt Momentum Funds.